

Track 43: Deep Learning & NLP Specialization

June 2026

Track 43: Deep Learning & NLP Specialization

Departments: AI & ML · AI & DS · CSE (elective) | MR ROKESH TECHNOLOGY

Skill path: Neural Networks → CNNs → RNNs/Transformers → NLP → Fine-tuning → Hugging Face → MLOps for DL

Related: [Track 05 — AI/ML](#) | [Track 17 — Prompt Engineering](#) | [Track 26 — Computer Vision](#)

Track Overview

Who this is for: AI & ML department students specializing in deep learning and natural language processing beyond foundational ML.

Prerequisites: Python, basic ML ([Track 5](#) M1–M3 or equivalent).

Tools: PyTorch, TensorFlow/Keras, Hugging Face Transformers, Weights & Biases, Jupyter, GPU (Colab/local).

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	DL awareness	Trained CNN + NLP classifier
1 Month	80–100	Junior DL engineer	Fine-tuned transformer model
3 Months	250–300	Internship-ready	Production NLP/CV capstone
6 Months	500–600	ML engineer entry	DL portfolio + deployment
9 Months	750–900	DL specialist	Multi-modal + LLM fine-tuning capstones

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Neural nets, backprop, PyTorch tensors	MNIST classifier
D3–4	CNNs: Conv layers, pooling, architectures	Image classifier
D5–6	RNNs, LSTM, text preprocessing	Sentiment classifier
D7–8	Transformers, attention, Hugging Face	BERT fine-tune demo
D9–10	Capstone	NLP or CV deep learning project

1-Month Foundation

Week	Topics	Project
W1	PyTorch deep dive, training loops, GPU	Custom CNN from scratch
W2	Transfer learning, ResNet, EfficientNet	Image classification app
W3	Transformers, tokenization, fine-tuning	Domain-specific text classifier
W4	Model evaluation, deployment (ONNX/Flask)	Deployed DL model API

3-Month Job Ready

Month	Topics	Projects
M1	Advanced CNNs, object detection (YOLO intro)	Object detector
M2	NLP: NER, summarization, RAG foundations	Document QA system
M3	MLOps for DL, experiment tracking, capstone	Production DL pipeline capstone

6-Month Professional

M1–M3 + LLM fine-tuning (LoRA) + multi-modal models + W&B monitoring + 3 deployed models.

9-Month Advanced

Capstones: Fine-tuned domain LLM + multi-modal AI application (text + image) | **Level:** L5

Assessment Rubric

Weekly Quiz 20% | Training labs 40% | Project 30% | Portfolio 10%

Appendix: Mentor Notes

Requires GPU access (Colab Pro or lab GPUs). Distinct from [Track 5](#) which covers broader ML; this track is DL/NLP depth for AI&ML department.